Do Now · retrieval

Networks, security and compression practice

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does DNS resolve?

Q2

What is a router's role?

Q3

What is lossy compression?

Q4

11010011_2 AND 00001111_2 as byte and hex?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Distinguish DNS addressing, IP routing, TCP transport and TLS security: each serves a different purpose.

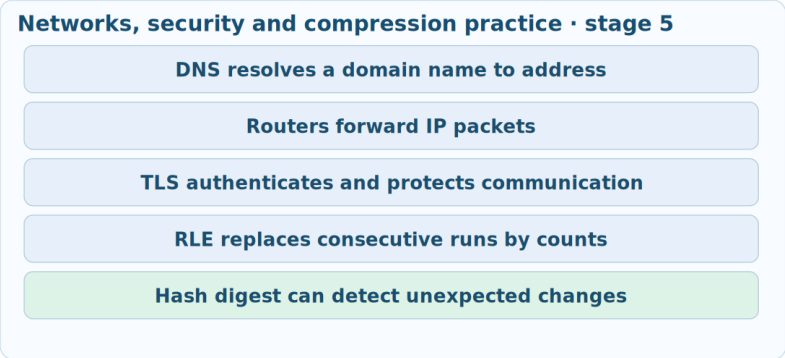
Core idea 2

Apply — RLE is lossless and suitable for runs; dictionary coding uses references. Lossy methods can discard detail.

Core idea 3

Evaluate — Hashing produces an integrity digest; symmetric and asymmetric encryption protect confidentiality using keys.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

DNS — Maps names to IP addresses.

TLS — Authenticated secure transport.

RLE — Run-length encoding.

digest — Output of hash function.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

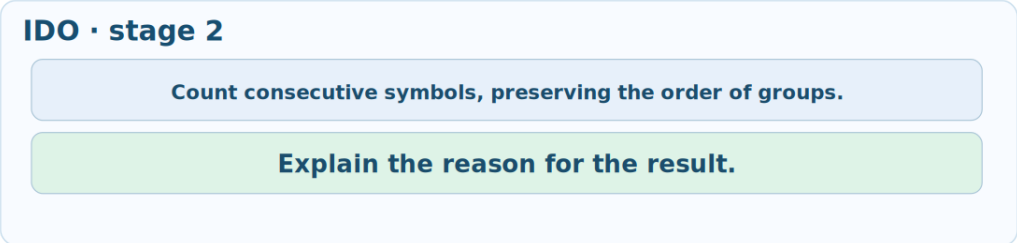
Problem

Encode AAABBCC with run-length encoding.

Follow the worked method

1. Predict first — Count consecutive symbols, preserving the order of groups.
2. Explain and check — Explain the reason for the result.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
A matching hash is evidence of integrity only when its expected value comes from a trustworthy channel.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Describe DNS, router and TLS roles when accessing HTTPS website. [6]

Question 2

RLE-encode AAAAABBBCCDD and explain how decoding works. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate transmitting official exam results as a lossy image versus an authenticated, encrypted, losslessly stored file. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Hashes can be decrypted using the public key to recover the input.

Use key vocabulary from memory

1. Explain DNS in your own words.

2. Explain TLS in your own words.

3. Explain RLE in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.